

Maximal Semantic Clique Enumeration in Vector-Attributed Graphs

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ABSTRACT

Large language model applications increasingly use GraphRAG to answer global questions over corpora whose evidence spans many documents. A typical pipeline constructs an entity–relation graph, organizes it through hierarchical community detection, and summarizes the resulting groups; group quality thus directly affects evidence coverage and coherence. Yet existing primitives do not yield groups that are both relation-complete and collectively coherent: community partitions allow missing links and suppress overlap, maximal cliques ignore vectors, and pairwise semantic thresholds fragment groups because of atypical pairs. We formulate *maximal semantic clique* (MSC) enumeration, which requires complete connectivity, constrains average pairwise vector similarity, and reports every inclusion-wise maximal qualifying clique. This aggregate constraint is non-monotone: an infeasible clique may have a feasible strict superset. Consequently, filtering only structural maximal cliques is incomplete, while standard Bron–Kerbosch pivoting and infeasibility pruning are unsound. We develop two complementary exact algorithms. MSC-BKPLUS combines structural maximal-clique enumeration with bounded semantic subset search. We prove that deterministically deleting a minimum-contribution vertex from a feasible clique of size at least three preserves feasibility. The theorem induces a canonical forest in which MSC-HIGHSIM generates every feasible clique exactly once without a visited-state table or materialized size layers. Across eight vector-attributed graphs and 24 workloads, at least one proposed method completes every workload, whereas the direct semantic Bron–Kerbosch baseline times out on 12. At q_{99} , MSC-HIGHSIM is $1.69\times$ – $6.24\times$ faster than MSC-BKPLUS. On three naturally attributed graphs, pairwise-threshold MCE attains only 0.156–0.556 Extended B^3 member recall, quantifying the group fragmentation avoided by the average constraint.

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The source code, data, and/or other artifacts have been made available at <https://github.com/CKTtari/maximal-semantic-clique-enumeration>.

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1 INTRODUCTION

Large language models increasingly answer questions over private, scientific, and enterprise corpora whose relevant evidence is distributed across many documents. Conventional vector retrieval is effective for local questions, but global questions require a system to identify recurring themes, connect multi-hop evidence, and summarize relationships at corpus scale. GraphRAG makes this group-discovery step explicit: it constructs an entity–relation graph, applies hierarchical community detection to organize closely related entities, and generates community summaries at multiple levels for global reasoning [15]. Related graph-based RAG systems expand and organize retrieved evidence through knowledge-graph relations [64]. Group quality therefore becomes an indexing concern, not merely a graph-analytics detail. An overly loose group produces an incoherent summary, an overly fragmented group misses distributed evidence, and a non-overlapping partition can hide entities that participate in multiple contexts. Modern AI applications consequently need compact, interpretable groups supported by both explicit relations and semantic signals.

Existing community and dense-pattern primitives do not provide such groups exactly. Hierarchical community detection is well suited to scalable, coarse-grained organization, but its communities may contain missing edges and its partition-oriented output can assign an entity to only one group at a given resolution. Query-oriented attributed-community search returns a community around a supplied seed rather than enumerating every qualifying group [20, 26, 47]. At the other extreme, maximal clique enumeration (MCE) guarantees that every pair of vertices is connected, but ignores vector attributes. Requiring every connected pair to exceed a semantic threshold is also brittle: a single atypical pair can split a group whose members remain coherent in aggregate. This is particularly problematic for cross-topic or cross-document evidence, where a semantically distant bridge may be essential even though the complete group remains structurally explicit and collectively meaningful.

We introduce *maximal semantic clique* (MSC) enumeration on a vector-attributed graph. Each vertex carries a real-valued vector, and a clique is semantically feasible when its average pairwise cosine similarity is at least a threshold τ . An MSC is an inclusion-wise maximal feasible clique: no feasible clique strictly contains it. The task is to enumerate all MSCs. Averaging captures collective semantic compatibility and tolerates a limited number of atypical pairs, while complete connectivity retains an unambiguous relation-level guarantee. Exact enumeration also preserves overlapping groups, allowing the same entity to support multiple retrieval or summarization contexts.

The aggregate semantic constraint fundamentally changes the enumeration problem. An MSC need not be a structural maximal clique, so enumerating structural maximal cliques and filtering them can miss valid answers. More importantly, average feasibility is non-monotone over clique containment: adding a vertex may raise or lower the average, and an infeasible clique may become feasible after further insertions. Pruning an infeasible partial clique can therefore discard a valid MSC. The structural argument underlying Bron–Kerbosch (BK) pivoting [5] also fails because a pivot may suppress a branch containing a semantic-maximal clique that cannot be safely extended by that pivot. We formalize these incompatibilities and prove that the associated size decision problem is NP-complete.

Our first algorithm, MSC-BKPLUS, restores structural pivoting by separating the structural and semantic searches. It first enumerates structural maximal cliques, using the fact that every MSC is contained in at least one such host. Within each host, it examines semantic subsets in decreasing size order. A sorted-edge upper bound and a vertex-contribution upper bound eliminate size layers that cannot contain a feasible subset, while local and global containment filters remove non-maximal and duplicate candidates. This decomposition amortizes semantic processing effectively when the threshold is permissive.

Our second algorithm, MSC-HIGHSIM, directly exploits the semantic threshold through canonical reverse search. For a feasible clique, define a vertex’s contribution as its total similarity to the other members. We prove that deleting a minimum-contribution vertex never decreases the remaining average below τ . Deterministic tie-breaking therefore assigns every feasible clique of size at least three a unique feasible parent, and repeated deletion reaches a feasible edge. The feasible cliques consequently form a canonical forest rooted at feasible edges. By reversing the parent relation, MSC-HIGHSIM reaches every feasible clique through feasible prefixes, rejects infeasible extensions and non-owned duplicate constructions before recursion, and requires neither a visited-state table nor materialized size layers. The two algorithms target complementary regimes: MSC-BKPLUS retains the benefits of structural decomposition at permissive thresholds, whereas MSC-HIGHSIM gains from contraction of the feasible state space at selective thresholds.

We make the following contributions.

Novel problem. We formulate exact MSC enumeration, a pattern-mining problem that combines complete connectivity with average pairwise semantic coherence while preserving overlapping, inclusion-wise maximal groups. We establish NP-completeness, prove that filtering structural maximal cliques is incomplete, and show why non-monotone average feasibility invalidates BK pivoting and conventional infeasibility pruning.

Exact algorithms. We develop two complementary exact algorithms with correctness guarantees. MSC-BKPLUS combines structural maximal-clique decomposition, two admissible upper bounds based on sorted edges and vertex contributions, and containment-aware subset enumeration. MSC-HIGHSIM follows from our feasible minimum-contribution deletion theorem, which exposes a unique-parent canonical forest inside the non-monotone search space. We prove that its reverse search generates every feasible clique exactly

once and returns exactly all MSCs without a visited-state table or materialized size layers.

Comprehensive evaluation. We evaluate exactness, end-to-end efficiency, memory consumption, threshold sensitivity, graph-size and parallel scalability, mechanism ablations, and output quality on three naturally attributed and five controlled graphs. At least one proposed method finishes all 24 primary workloads, whereas MSC-BKBASIC times out on 12. At q_{99} , MSC-HIGHSIM is $1.69\times$ – $6.24\times$ faster than MSC-BKPLUS. Across the four-graph ablation panel, canonical ownership is $3.64\times$ – $4.38\times$ faster and uses $1.53\times$ – $1.73\times$ less peak memory than duplicate-state organizations. Pairwise-threshold member recall is 0.156 – 0.556 on the three naturally attributed graphs.

2 PRELIMINARIES

2.1 Notations and Problem Definition

We study a *vector-attributed graph* $\mathcal{G} = (V, E, X)$, where V is the vertex set, $E \subseteq \binom{V}{2}$ is the edge set, and $X = \{\mathbf{x}_v \in \mathbb{R}^D \mid v \in V\}$ assigns a D -dimensional attribute vector to each vertex.

DEFINITION 1 (SIMILARITY FUNCTION). *We instantiate semantic similarity by cosine similarity. After ℓ_2 normalization, $\text{sim}(u, v) = \mathbf{x}_u^\top \mathbf{x}_v$ and $\text{sim}(u, v) \in [-1, 1]$.*

Throughout this paper we work with normalized vectors and precompute $\text{sim}(u, v)$ for all $(u, v) \in E$. The threshold $\tau \in [-1, 1]$ is drawn from the same range.

DEFINITION 2 (CLIQUE). *A set $C \subseteq V$ is a clique in $\mathcal{G}$ if every pair of distinct vertices in C is connected by an edge, i.e., $\{u, v\} \in E$ for all $u \neq v \in C$.*

Classical MCE seeks cliques that cannot be extended while preserving complete connectivity. In a vector-attributed graph, the attributes add a second, group-level question: how similar are the members represented by a structural clique? We express this semantic condition through two companion quantities.

DEFINITION 3 (PAIRWISE SIMILARITY SUM AND AVERAGE PAIRWISE SIMILARITY). *For a clique C of size $k = |C| \geq 2$, the pairwise similarity sum is*

$$\text{simSum}(C) = \sum_{\{u,v\} \subseteq C} \text{sim}(u, v),$$

and the average pairwise similarity is

$$\text{avgSim}(C) = \frac{\text{simSum}(C)}{\binom{k}{2}}.$$

DEFINITION 4 (SEMANTIC CLIQUE). *Given a threshold $\tau \in [-1, 1]$, a clique C with $|C| \geq 2$ is a τ -semantic clique if $\text{avgSim}(C) \geq \tau$.*

For algorithmic purposes, we equivalently track the *semantic surplus*

$$W(C) = \text{simSum}(C) - \tau \binom{k}{2},$$

which is non-negative if and only if $\text{avgSim}(C) \geq \tau$. Working with simSum and W avoids division at every recursive step; avgSim is used only when a human-readable quantity is required.

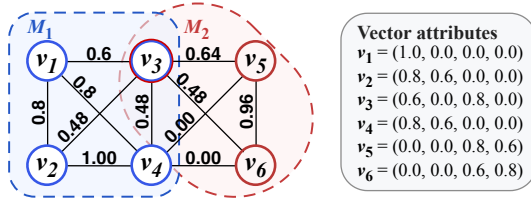


Figure 1: Running example with vector attributes and edge similarities.

DEFINITION 5 (MAXIMAL SEMANTIC CLIQUE (MSC)). Given $\mathcal{G} = (V, E, X)$ and threshold $\tau \in [-1, 1]$, a τ -semantic clique $C \subseteq V$ is a maximal semantic clique (MSC) if there is no τ -semantic clique $C' \supsetneq C$.

Inclusion-wise maximality requires considering all strict supersets; the absence of a feasible single-vertex extension alone is insufficient because average feasibility is non-monotone. For example, let $\tau = 0.6$ and let four vertices form a clique with $\text{sim}(a, b) = 0.6$, $\text{sim}(a, u) = \text{sim}(b, u) = \text{sim}(a, v) = \text{sim}(b, v) = 0.5$, and $\text{sim}(u, v) = 1$. The feasible clique $\{a, b\}$ has no feasible single-vertex extension, since either three-vertex average is $1.6/3 < 0.6$, but the strict superset $\{a, b, u, v\}$ is feasible with average $3.6/6 = 0.6$. Consequently, algorithms may use the absence of a feasible single extension only to generate candidates; a global containment test is still required for MSC output.

EXAMPLE 1. Figure 1 shows a vector-attributed graph with six vertices. Edge labels give the precomputed pairwise similarities. With threshold $\tau = 0.6$, consider the two structural cliques

$$M_1 = \{v_1, v_2, v_3, v_4\}, \quad M_2 = \{v_3, v_4, v_5, v_6\}.$$

Let $C_1 = \{v_1, v_2, v_3, v_4\}$, $C_2 = \{v_3, v_5, v_6\}$, and $C_3 = \{v_3, v_4, v_5, v_6\}$. Their total and average similarities are

$$\begin{aligned} \text{simSum}(C_1) &= 4.16, & \text{avgSim}(C_1) &\approx 0.693 \geq \tau, \\ \text{simSum}(C_2) &= 2.08, & \text{avgSim}(C_2) &\approx 0.693 \geq \tau, \\ \text{simSum}(C_3) &= 2.56, & \text{avgSim}(C_3) &\approx 0.427 < \tau. \end{aligned}$$

Thus C_1 and C_2 are τ -semantic cliques, whereas the structurally valid clique C_3 is not.

Moreover, neither C_1 nor C_2 has a strict superset that is a τ -semantic clique, so both are MSCs.

Problem Statement.

PROBLEM 1 (MSC ENUMERATION). Given a vector-attributed graph $\mathcal{G} = (V, E, X)$ and a threshold $\tau \in [-1, 1]$, enumerate all MSCs of $\mathcal{G}$.

2.2 Challenges

HARDNESS. MSC enumeration contains classical MCE as a special case. When singleton outputs are excluded and τ does not exceed the minimum similarity on any edge, every structural maximal clique of size at least two satisfies the semantic constraint, so the $\Omega(3^{n/3})$ output lower bound carries over. The associated decision problem is NP-complete under finite-precision rational similarities and thresholds of polynomial encoding length.

THEOREM 2.1. Given $\mathcal{G} = (V, E, X)$, threshold $\tau \in [-1, 1]$, and integer k , deciding whether an MSC of size $\geq k$ exists is NP-complete.

PROOF. NP membership. The decision instance is positive if and only if it contains a τ -semantic clique C of size at least k . The forward implication is immediate. For the reverse implication, the finite family of feasible clique supersets of C contains an inclusion-wise maximal member, whose size is at least $|C| \geq k$. A certificate therefore need only provide such a feasible clique C : verifying all edges and $\text{avgSim}(C) \geq \tau$ takes $O(|C|^2)$ time. Accordingly, NP verification requires only this feasibility certificate. If maximality were tested separately, examining only single-vertex extensions would be insufficient under the non-monotone constraint. **NP-hardness.** We reduce from the CLIQUE decision problem [28], which asks whether $G_0 = (V, E)$ contains a clique of size $\geq k$. Assign $\mathbf{x}_v = \mathbf{e}_1$ to every vertex, so $\text{sim}(u, v) = 1$ for every edge, and set $\tau = 1$. Under this assignment $\text{avgSim}(C) = 1 \geq \tau$ for every clique C , so every structural maximal clique is an MSC and vice versa. Any clique of size $\geq k$ in G_0 is contained in some structural maximal clique of size $\geq k$, which is also an MSC of size $\geq k$. Conversely, any MSC of size $\geq k$ is a clique of size $\geq k$ in G_0 . Hence a size- k MSC exists if and only if G_0 contains a clique of size $\geq k$, completing the reduction. $\square$

The output sets of MSC enumeration and MCE are incomparable: a structural maximal clique with $\text{avgSim}(M) < \tau$ may contain semantically maximal subsets that MCE never generates, while valid MSCs need not be structurally maximal. Therefore, enumerating structural maximal cliques and retaining only those that pass the average constraint is incomplete because an MSC need not be structurally maximal. The output size, and hence the computational cost, follows the threshold-dependent behavior characterized below.

Output Distribution and Practical Scope. The number and sizes of MSCs vary substantially with τ . At low thresholds, the output approaches classical MCE. As τ increases, infeasible structural cliques can fragment into many incomparable feasible sub-cliques, inflating both the output and the unavoidable enumeration cost. At selective thresholds, most cliques become infeasible and the output contracts. Because these transitions depend on each dataset's similarity distribution, Section 5 uses a fixed grid of edge-similarity quantiles rather than runtime-selected absolute thresholds.

Non-monotonicity and Framework Incompatibility. The core difficulty is that avgSim is non-monotone over the subset lattice. In classical MCE, structural feasibility is monotone and maximality is enforced implicitly by the $P = \emptyset$ invariant at every leaf, so the set of feasible extensions can be characterized solely by the current state of C . Under the average constraint, no such invariant exists: a partial clique with $\text{avgSim}(C) < \tau$ may still grow into a valid MSC, while one with $\text{avgSim}(C) \geq \tau$ may admit no feasible extension. Furthermore, whether a candidate vertex can participate in a valid extension depends on its accumulated similarity to the current partial clique, which evolves throughout the recursion and cannot be captured by structural conditions alone. As a result, the pruning mechanisms based on structural monotonicity lose their foundation, and the pivot strategy no longer applies. A direct BK adaptation must therefore retain structurally valid intermediate states even when they are currently infeasible. In the baseline developed next,

Algorithm 1: MSC-BKBASIC

```

1 Input:  $G = (V, E)$ , semantic vectors  $\{x_v\}$ , threshold  $\tau$ 
2 Output: All MSCs  $\mathcal{R}$ 
3 Precompute  $\text{sim}(u, v)$  and degeneracy ordering  $\sigma$ ;
4 Each worker initializes a private result set  $\mathcal{R}_t \leftarrow \emptyset$ ;
5 for each  $v \in \sigma$  in parallel do
6    $P \leftarrow \{u \in N(v) \mid \sigma(u) > \sigma(v)\}$ ;
7    $X \leftarrow \{u \in N(v) \mid \sigma(u) < \sigma(v)\}$ ;
8   Search( $C = \{v\}, s_C = 0, P, X, \text{mode} = \text{ENUM}, 0$ )
9 return  $\bigcup_t \mathcal{R}_t$ ;

10 Function Search( $C, s_C, P, X, \text{mode}, k_{\min}$ ):
11    $k \leftarrow |C|$ ,  $\text{hasFeasibleDesc} \leftarrow \text{false}$ ;
12    $Q \leftarrow P$ ;
13   for each  $v \in Q$  do
14      $s_D \leftarrow s_C + \sum_{u \in C} \text{sim}(u, v)$ ;
15      $P \leftarrow P \setminus \{v\}, P' \leftarrow P \cap N(v), X' \leftarrow X \cap N(v)$ ;
16      $\text{res} \leftarrow \text{Search}(C \cup \{v\}, s_D, P', X', \text{mode}, k_{\min})$ ;
17     if  $\text{res}$  then
18        $\text{hasFeasibleDesc} \leftarrow \text{true}$ ;
19       if  $\text{mode} = \text{CHECK}$  then return true;
20    $X \leftarrow X \cup \{v\}$ ;
21 if  $\neg \text{hasFeasibleDesc}$  then
22   if  $k \geq 2 \wedge s_C < \tau \binom{k}{2}$  then return false;
23   if  $\text{mode} = \text{CHECK}$  then return  $k > k_{\min}$ ;
24   if  $X \neq \emptyset$  then
25     if Search( $C, s_C, X, \emptyset, \text{CHECK}, k$ ) then
26       return true;
27   if  $k \geq 2$  then  $\mathcal{R}_t \leftarrow \mathcal{R}_t \cup \{C\}$ ;
28   return true;
29 return  $\text{hasFeasibleDesc}$ ;

```

τ validates terminal semantic candidates but does not reduce the structural recursion itself.

3 FRAMEWORK

Framework Overview. MSC-BKBASIC is a direct BK baseline that retains degeneracy-ordered (P, X) recursion while omitting the structurally justified pivot. At each leaf, a secondary CHECK pass over X enforces inclusion-wise semantic maximality.

Figure 2 exposes the graph states behind both terminal decisions. In Figure 2(a), the feasible terminal $D = \{v_1, v_3, v_4\}$ has $X = \{v_2\}$; CHECK reconstructs the feasible strict superset C_1 and suppresses D . Figure 2(b) follows the v_6 -owned branch from $C = \{v_6, v_3\}$. The first insertion, v_4 , remains infeasible even after adding v_5 . After backtracking, inserting v_5 yields C_2 with v_4 in its exclusion set. CHECK reaches infeasible C_3 , returns false, and the outer search emits C_2 .

Degeneracy-Root Ownership. Degeneracy ordering assigns each clique to its earliest vertex under σ : later neighbors form P , and earlier neighbors form X . This unique seed owns every clique, including each MSC, while bounding the initial candidate set by graph degeneracy.

Emit a feasible terminal only if CHECK finds no feasible strict superset in X .

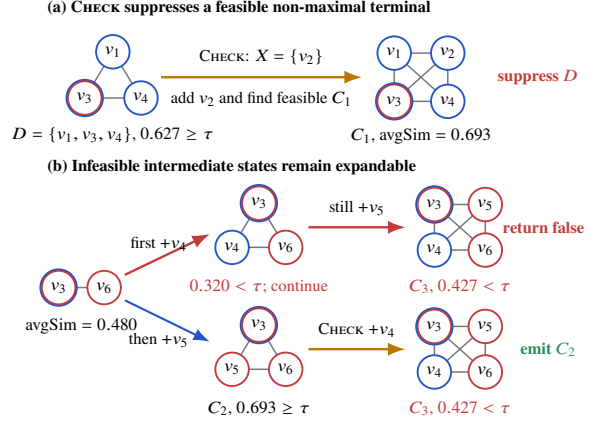


Figure 2: Selected MSC-BKBASIC branches on the running example.

Terminal Feasibility and Maximality Check. The Boolean return value records whether a subtree contains a feasible terminal. Only when no child reports one does C become a candidate; an infeasible candidate is discarded. A feasible candidate can still have a strict extension through X , so CHECK searches that exclusion set and suppresses C upon finding one. The parameter $k_{\min} = |C|$ enforces strict containment because the check recursion may also revisit sets no larger than C .

Why pivoting is not used. In standard BK, a pivot $p \in P \cup X$ guarantees that each structural maximal clique in the current branch contains either p or a vertex in $P \setminus N(p)$. Hence vertices in $P \cap N(p)$ need not become expansion roots. This argument depends on structural feasibility being preserved when an adjacent pivot is added.

The average constraint invalidates that implication. Consider a complete graph on $\{s, p, u, w\}$ at the BK state $C = \{s\}$, $P = \{p, u, w\}$, and $X = \emptyset$. Let the three similarities within $\{s, u, w\}$ equal 0.9, the three similarities incident to p equal 0, and $\tau = 0.6$. Then $\{s, u, w\}$ is an MSC with average 0.9, whereas its only strict clique superset has average 0.45. Choosing p as the pivot leaves only p in $P \setminus N(p)$, so every explored child contains p and the valid MSC $\{s, u, w\}$ is lost. Pivot-free branching is therefore required for the direct semantic BK baseline.

Complexity.

THEOREM 3.1 (TIME AND SPACE COMPLEXITY). *Let d_G be the degeneracy of G , let $\mathcal{L}$ be the feasible terminal states reached by the outer search, and let X_C be the candidate set examined by CHECK for $C \in \mathcal{L}$. The running time is $O^*(n2^{d_G} + \sum_{C \in \mathcal{L}} 2^{|X_C|})$, where O^* suppresses polynomial set-intersection costs. A graph-independent upper bound is $O^*(3^n)$. With T workers, the implementation uses $O(m + nD + Tn^2 + \sum_{C \in \mathcal{R}} |C|)$ worst-case space.*

PROOF. A degeneracy seed has at most d_G later neighbors. Without pivoting, its outer recursion may visit every subset of those neighbors, giving $O^*(n2^{d_G})$ states. For a feasible terminal C , CHECK

may independently visit every subset of X_C . The same earlier extension can be reconsidered for several terminals; charging a state to a pair of nested vertex sets gives the safe $O^*(3^n)$ bound. Similarities and sparse adjacency and stored outputs require $O(m + nD + \sum_{C \in \mathcal{R}} |C|)$ storage. Each worker owns an $O(n)$ accumulator, while decreasing candidate arrays retained by a worst-case CHECK stack total $O(n^2)$. $\square$

For comparison, degeneracy-oriented pivoted MCE has the time bound [16, 17]

$$O(d_G n 3^{d_G/3})$$

The average constraint introduces more than an $O(|C|^2)$ validation cost: valid MSCs may occur at nonmaximal structural states, so the direct search may inspect all 2^{d_G} subsets below a seed and revisit earlier-neighbor subsets in CHECK. This gap formalizes the additional search complexity relative to structural MCE.

Design Implication. MSC-BKPLUS is a direct same-definition baseline that uses local search state and no global visited table. Its pivot-free main search cannot prune every infeasible intermediate, while CHECK may revisit earlier-neighbor subspaces. Section 4 addresses these costs with two search organizations tailored to different threshold regimes.

4 PROPOSED ALGORITHMS

This section presents two exact algorithms for complementary threshold regimes. MSC-BKPLUS decouples structural enumeration from semantic subset processing, retaining Tomita pivoting and using admissible size-layer bounds. MSC-HIGHSIM uses canonical reverse search, allowing τ to reject infeasible child states before recursion. We evaluate both methods on the same quantile grid: MSC-BKPLUS provides broad threshold coverage, while MSC-HIGHSIM targets selective high- τ workloads.

4.1 MSC-BKPLUS: Structural Decomposition and Subset Enumeration

Structural-Semantic Decoupling. MSC-BKPLUS restores the Tomita pivot by separating structural enumeration from semantic subset search. Pivoted BK first produces structural maximal cliques; the average constraint is then evaluated only within each resulting host. Lemma 4.1 establishes that this separation preserves every MSC.

For a host clique M and target size k , let $\bar{e}_k$ be the mean of the $\binom{k}{2}$ largest pairwise similarities in M . For each $v \in M$, let $q_v(k)$ be the sum of its $k-1$ largest incident similarities within M , and let $\bar{q}_k$ be the sum of the k largest $q_v(k)$ values divided by $k(k-1)$. Both quantities upper-bound the average similarity of any size- k subset; Properties 4.1 and 4.2 prove these bounds below.

Figure 3(a) first exposes the two overlapping structural hosts $M_1 = \{v_1, v_2, v_3, v_4\}$ and $M_2 = \{v_3, v_4, v_5, v_6\}$. The semantic phase handles each host independently. Since M_1 satisfies the average constraint, Figure 3(b) retains it directly as C_1 . The infeasible M_2 instead activates the bounded subset search in Figure 3(c), which recovers the feasible triangle $C_2 = \{v_3, v_5, v_6\}$. The global containment filter then removes duplicate candidates and strict subsets before returning $\mathcal{R} = \{C_1, C_2\}$ (Figure 3(d)).

Algorithm 2: MSC-BKPLUS

Input: $G = (V, E)$, similarities $\{\text{sim}(u, v)\}_{(u,v) \in E}$, threshold τ
Output: All MSCs $\mathcal{R}$

```

1  $\mathcal{F} \leftarrow \emptyset$ ;
   /* Pipelined structural MCE and semantic processing */
2 Build degeneracy ordering  $\sigma$ ;
3 for each  $v \in \sigma$  in parallel do
4    $P \leftarrow \{u \in N(v) \mid \sigma(u) > \sigma(v)\}$ ;
5    $X \leftarrow \{u \in N(v) \mid \sigma(u) < \sigma(v)\}$ ;
6   Run BK( $\{v\}, P, X$ ) with the following callback at each leaf  $M$ ;
   /* Callback: intra-clique semantic enumeration */
7 for each structural leaf  $M$  produced by a worker do
8   if  $\text{avgSim}(M) \geq \tau$  then  $\mathcal{F} \leftarrow \mathcal{F} \cup \{M\}$ ; continue;
   /* Admissible layer bounds */
9   Sort edges of  $M$  by similarity descending;
10   $k^* \leftarrow \max\{k \in [2, |M| - 1] \mid \bar{e}_k \geq \tau\}$ ;
11  if  $k^*$  does not exist then continue;
12   $lmax \leftarrow \emptyset$ ;  $umask \leftarrow \emptyset$ ;
13  for  $k \leftarrow k^*$  to 2 do
14    Compute the vertex-contribution bound  $\bar{q}_k$  for size  $k$ ;
15    if  $\bar{q}_k < \tau$  then continue;
16    for each size- $k$  subset  $S \subseteq M$  do
17      if  $(S \& umask) = S$  and  $\exists T \in lmax : S \subseteq T$  then
18        continue;
19      if  $\text{avgSim}(S) \geq \tau$  then
20         $lmax \leftarrow lmax \cup \{S\}$ ;
21         $umask \leftarrow umask \mid S$ ;
22         $\mathcal{F} \leftarrow \mathcal{F} \cup \{S\}$ ;
   /* Phase 3: Exact global filter */
23 Sort  $\mathcal{F}$  by  $|\cdot|$  descending; remove exact duplicates;
24 Initialize an empty inverted index  $inv : v \mapsto \emptyset$ ;
25  $\mathcal{R} \leftarrow \emptyset$ ;
26 for each equal-size group  $\mathcal{G} \subseteq \mathcal{F}$  in descending size do
27   for each  $S \in \mathcal{G}$  in parallel do
28      $r \leftarrow \arg \min_{v \in S} |inv[v]|$ ;
29     Keep  $S$  iff no  $T$  referenced by  $inv[r]$  satisfies  $S \subseteq T$ ;
30 Insert every kept  $S \in \mathcal{G}$  into  $\mathcal{R}$  and  $inv$ ;
31 return  $\mathcal{R}$ ;
```

LEMMA 4.1 (STRUCTURAL SUPERSET OF MSCs). *Every MSC is contained in at least one structural maximal clique of G . That is, for every MSC C , there exists a structural maximal clique M such that $C \subseteq M$.*

PROOF. By definition, every MSC is a clique of G . Let C be an arbitrary MSC, and consider the family

$$\mathcal{F} = \{D \subseteq V \mid C \subseteq D \text{ and } D \text{ is a clique in } G\}.$$

Since $C \in \mathcal{F}$, the family is nonempty. Because V is finite, $\mathcal{F}$ is finite, so we may choose a set $M \in \mathcal{F}$ that is maximal under inclusion. By construction, M is a clique containing C , and no vertex $w \in V \setminus M$ extends M to a larger clique; hence M is a structural maximal clique of G , and $C \subseteq M$. $\square$

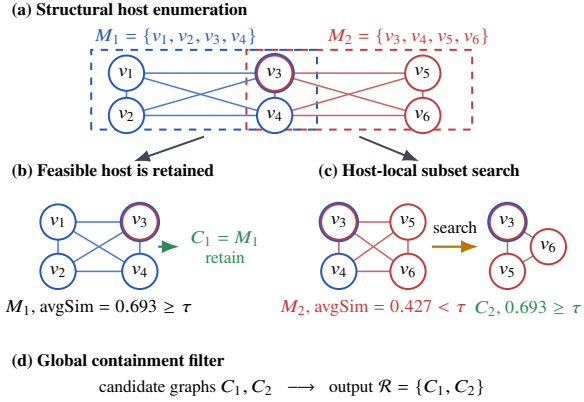


Figure 3: MSC-BKPLUS execution on the running example.

Structural Host Enumeration. Phase 1 uses degeneracy-oriented, pivoted BK [17, 51] and is independent of τ . Degeneracy roots are dynamically scheduled; upon reaching a structural leaf M , the owning worker immediately invokes the semantic callback instead of materializing the complete family of structural maximal cliques.

Bounded Semantic Subset Search. For each host M , Phase 2 visits subset sizes in descending order and uses a host-local similarity matrix for repeated feasibility tests. The local filter stores previously found feasible sets in $lmax$ and uses $umask = \bigcup_{T \in lmax} T$ as a fast prerequisite for an exact containment probe. If $S \subseteq T$ for feasible T , then S cannot be an MSC; the filter therefore removes S without changing the output. The admissible bounds below skip entire size layers before subset generation.

Global Maximality Reconciliation. An MSC may occur in multiple structural maximal cliques, so semantic subset search may emit duplicate or subsumed candidates. Global reconciliation consolidates them into $\mathcal{R}$. Candidates are deduplicated and processed by decreasing size. For each S , the inverted index is probed through its rarest vertex, limiting exact containment tests to retained supersets that share that vertex. Equal-size queries read a frozen index in parallel and are inserted only after the group completes. Because equal-size sets cannot strictly contain one another and discarded supersets are covered transitively by retained ones, this grouped procedure is equivalent to testing all larger candidates.

Admissible Size-Layer Bounds. Phase 2 as described iterates over all subset sizes from $|M| - 1$ down to 2. A key observation allows the starting size to be reduced: the best achievable avgSim for any size- k subset of M is bounded above by the mean of the $\binom{k}{2}$ largest pairwise similarities in M .

PROPERTY 4.1 (SORTED-EDGE UPPER BOUND). Let $e_1 \geq e_2 \geq \dots \geq e_{\binom{m}{2}}$ be the pairwise similarities of M sorted in descending order. For any size- k subset $S \subseteq M$,

$$\text{avgSim}(S) \leq \bar{e}_k := \frac{1}{\binom{k}{2}} \sum_{i=1}^{\binom{k}{2}} e_i.$$

PROOF. Any size- k subset $S \subseteq M$ uses exactly $\binom{k}{2}$ pairs, each contributing at most the corresponding value in the sorted sequence

$e_1 \geq e_2 \geq \dots$; hence $\text{simSum}(S) \leq \sum_{i=1}^{\binom{k}{2}} e_i$, and dividing both sides by $\binom{k}{2}$ gives the result. $\square$

The sorted-edge bound ignores that its largest edges may be concentrated on too few vertices to coexist in one size- k subset. A complementary bound tracks which vertices support those similarities. For each $v \in M$, let $q_v(k)$ be the sum of the $k - 1$ largest similarities between v and vertices in $M \setminus \{v\}$. Let $q_{(1)}(k) \geq \dots \geq q_{(m)}(k)$ denote these values in descending order.

PROPERTY 4.2 (VERTEX-CONTRIBUTION UPPER BOUND). For any size- k subset $S \subseteq M$,

$$\text{avgSim}(S) \leq \bar{q}_k := \frac{1}{k(k-1)} \sum_{i=1}^k q_{(i)}(k).$$

PROOF. For every $v \in S$, the sum of its similarities to the other vertices of S uses $k - 1$ vertices from $M \setminus \{v\}$ and is therefore at most $q_v(k)$. Summing over S counts each internal pair twice, so

$$2\text{simSum}(S) \leq \sum_{v \in S} q_v(k) \leq \sum_{i=1}^k q_{(i)}(k).$$

Division by $k(k-1)$ gives the claimed average-similarity bound. $\square$

The prefix means $\bar{e}_k$ are non-increasing in k . The algorithm identifies

$$k^* = \max\{k \in [2, |M| - 1] \mid \bar{e}_k \geq \tau\},$$

and skips every larger layer. For each remaining k , it also skips the layer when $\bar{q}_k < \tau$. Both bounds are admissible independently, so their joint use cannot remove a feasible subset. Sorting the edge and per-vertex similarity lists costs $O(|M|^2 \log |M|)$ per host.

Applicable Regime. MSC-BKPLUS applies Tomita pivoting during structural MCE, whose cost is $O(d_G n 3^{d_G/3})$ and independent of τ . At low thresholds, large feasible subsets are identified early within each structural host. At higher thresholds, the sorted-edge and vertex-contribution bounds skip infeasible size layers. Structural MCE remains a fixed phase cost, which motivates the canonical search developed next for selective thresholds.

Complexity.

THEOREM 4.1 (TIME COMPLEXITY OF MSC-BKPLUS). Let $\mathcal{M}$ denote the family of all structural maximal cliques of G . For each $M \in \mathcal{M}$, let Q_M denote the number of exact local containment probes and let $b_M = \lceil |M|/w \rceil$ be its bitmap length in machine words. MSC-BKPLUS runs in

$$O\left(d_G n 3^{d_G/3} + \sum_{M \in \mathcal{M}} \left(2^{|M|} |M|^2 + Q_M b_M\right) + KN \log N + K \sum_{S \in \mathcal{F}} L_{\min}(S)\right)$$

time, where d_G is the degeneracy of G , $N = |\mathcal{F}|$ is the number of candidates before global deduplication, $K = \max_{S \in \mathcal{F}} |S|$, and $L_{\min}(S) = \min_{v \in S} |\text{inv}[v]|$.

PROOF. The structural phase uses degeneracy-oriented Tomita BK [17], with cost $O(d_G n 3^{d_G/3})$. For a host M of size h , subset enumeration visits at most 2^h subsets and evaluates each from the cached similarity matrix in $O(h^2)$ time. Exact local maximality performs Q_M bitmap containment probes of b_M words each; keeping this term explicit also covers instances with many feasible local subsets. The final filter sorts N candidates in $O(KN \log N)$ time. It

then scans the shortest inverted list for each S , with at most $L_{\min}(S)$ sorted-set containment checks of $O(K)$ time. $\square$

Within a host, non-monotone feasibility retains the exponential worst case; the admissible bounds and local containment reduce inspected candidates without changing the output.

4.2 MSC-HIGHSIM: High- τ Canonical Reverse Search

Motivation and Core Insight. Even with the size-layer bounds, MSC-BKPLUS is bound to Phase 1: it enumerates all structural maximal cliques before any semantic evaluation takes place. In the high- τ regime, the vast majority of structural cliques may contain no feasible semantic result, yet each still contributes Phase 1 cost. More fundamentally, the BK framework offers no mechanism to incorporate τ into branching decisions—the threshold appears only in validation, never in the search logic itself. This structural overhead motivates abandoning the BK skeleton entirely. We call the resulting method MSC-HIGHSIM because increasing τ directly contracts its feasible canonical-state space, making selective high- τ workloads its intended operating regime.

The obstacle to integrating τ directly into search is the non-monotonicity of avgSim: an infeasible clique can have a feasible strict superset. A search that retains only feasible states is therefore complete only if every feasible target has a feasible construction path. Theorem 4.2 supplies such a path through the deletion rule defined next.

For a vertex v in a clique C , define its *similarity contribution* as $a_C(v) = \sum_{u \in C \setminus \{v\}} \text{sim}(u, v)$: the sum of its similarities to the other vertices in C . We delete a vertex with minimum similarity contribution, breaking ties by vertex identifier. The theorem below proves that the remaining clique is feasible; the deterministic deletion therefore defines the canonical parent used by reverse search.

The implementation traverses canonical trees by DFS from feasible edge roots. It stores no complete size layer; a final containment filter enforces inclusion-wise maximality.

Feasible Parents and the Canonical Forest.

THEOREM 4.2 (FEASIBLE MINIMUM-CONTRIBUTION DELETION). *Let C be a τ -semantic clique of size $k \geq 3$, with the similarity contribution $a_C(v)$ defined above. If $d(C)$ minimizes the pair $(a_C(v), v)$ lexicographically, then $C \setminus \{d(C)\}$ is a τ -semantic clique. Repeated application of this rule therefore reaches a feasible edge, and reversing these deletions gives a non-increasing-average insertion path.*

PROOF. Each pair is counted once at each endpoint. The minimum contribution therefore satisfies $a_C(d(C)) \leq 2\text{simSum}(C)/k$. Deleting $d(C)$ leaves similarity sum at least $\text{simSum}(C)(k-2)/k$, and hence

$$\text{avgSim}(C \setminus \{d(C)\}) \geq \frac{\text{simSum}(C)(k-2)/k}{\binom{k-1}{2}} = \text{avgSim}(C) \geq \tau.$$

Deterministic tie-breaking makes $d(C)$ unique. Applying the same argument until two vertices remain preserves feasibility at every deletion; reversing the sequence proves the insertion-path claim. $\square$

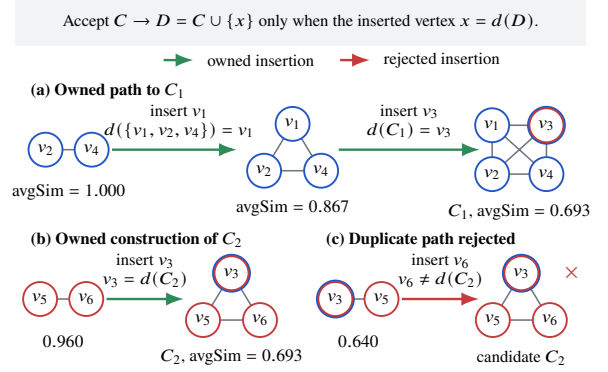


Figure 4: Canonical ownership on the running example.

Define $p(C) = C \setminus \{d(C)\}$ for every feasible clique C of size at least three. The theorem makes $p(C)$ a feasible and unique parent, so all feasible cliques form a forest rooted at feasible edges. This property does not make arbitrary insertions or deletions monotone; it identifies one complete feasible path for every target. MSC-HIGHSIM traverses precisely these owned parent-child edges and rejects alternative constructions before recursion.

Figure 4 applies this rule to the running example. The upper sequence constructs C_1 through canonical insertions. The lower panels contrast the owned construction of C_2 with an alternative insertion that forms the same clique but is rejected before recursion because the inserted vertex is not $d(C_2)$.

Feasible Edge Roots. Every feasible clique of size at least two reaches a feasible edge under the deterministic deletion rule, so Algorithm 3 starts one tree for each edge with similarity at least τ . A state stores the clique C , its pairwise sum $s_C = \text{simSum}(C)$, the common-neighbor set P , candidate accumulators acc , and the member contributions a_C . No BK exclusion set is required: uniqueness follows from the canonical-parent test rather than from an insertion order.

Surplus and Canonical-Child Tests. For $k = |C|$ and $\delta = acc[v]$, the extension satisfies

$$W(C \cup \{v\}) = W(C) + \delta - \tau k.$$

If this value is negative, the child itself is infeasible and cannot lie on the canonical path of any feasible answer, so it is safely skipped. For a feasible child $D = C \cup \{v\}$, the algorithm updates every member's similarity contribution and accepts D only if $v = d(D)$. Equivalently, the canonical parent of D must be exactly C . This test removes alternative insertion paths before recursion rather than after state materialization. The flag *hasValid* is set before the ownership test because it records the existence of any feasible one-vertex extension, not the existence of an owned child. A state with such an extension cannot itself be inclusion-wise maximal; the extension is generated from its unique canonical parent.

The canonical test removes duplicate insertion paths before recursion. Accumulator values are restored during backtracking, so the result does not depend on the order of floating-point additions.

Inclusion-Wise Maximality. A state with no feasible single-vertex extension is added to $\mathcal{F}$, but is not yet declared an MSC. If

Algorithm 3: MSC-HIGHSIM

```

1 Input:  $G = (V, E)$ , semantic vectors  $\{x_v\}$ , threshold  $\tau$ 
2 Output: All MSCs  $\mathcal{R}$ 
3 Precompute  $\text{sim}(u, v)$  for all  $(u, v) \in E$ ;
4  $\mathcal{F} \leftarrow \emptyset$ ;
5 for each  $(u, v) \in E$  with  $u < v$  and  $\text{sim}(u, v) \geq \tau$  in parallel do
6    $C \leftarrow \{u, v\}$ ;  $s_C \leftarrow \text{sim}(u, v)$ ;
7    $P \leftarrow N(u) \cap N(v)$ ;
8    $\text{acc}[w] \leftarrow \text{sim}(w, u) + \text{sim}(w, v)$  for  $w \in P$ ;
9    $a_C(u) = a_C(v) \leftarrow \text{sim}(u, v)$ ;
10  Search( $C, s_C, P, \text{acc}, a_C$ );
11 Apply Algorithm 2's grouped strict-containment filter to  $\mathcal{F}$ ,
   obtaining  $\mathcal{R}$ ;
12 return  $\mathcal{R}$ ;

13 Function Search( $C, s_C, P, \text{acc}, a_C$ ):
14    $k \leftarrow |C|$ ;  $w_C \leftarrow s_C - \tau \binom{k}{2}$ ;
15    $\text{hasValid} \leftarrow \text{false}$ ;
16   for each  $v \in P$  do
17      $\delta \leftarrow \text{acc}[v]$ ;
18     if  $w_C + \delta - \tau k < 0$  then continue;
19      $\text{hasValid} \leftarrow \text{true}$ ;
20      $D \leftarrow C \cup \{v\}$ ;
21     Update  $a_D$  from  $a_C$  and  $\text{sim}(v, u)$  for  $u \in C$ ;
22     if  $v \neq \arg \min_{x \in D} (a_D(x), x)$  then continue;
23      $P' \leftarrow P \cap N(v)$ ;
24      $\text{acc}'[w] \leftarrow \text{acc}[w] + \text{sim}(w, v)$  for  $w \in P'$ ;
25     Search( $D, s_C + \delta, P', \text{acc}', a_D$ );
26   if not  $\text{hasValid}$  then  $\mathcal{F} \leftarrow \mathcal{F} \cup \{C\}$ ;

```

such a candidate C is not inclusion-wise maximal, the finite family of feasible clique supersets of C contains an inclusion-wise maximal member M . The canonical search enumerates M exactly once, and M has no feasible single extension, so it also belongs to $\mathcal{F}$. The final grouped inverted-index containment filter consequently removes C . Conversely, an MSC is not contained in any other feasible candidate and is retained.

THEOREM 4.3 (ENUMERATION CORRECTNESS). *Algorithm 3 generates every τ -semantic clique of size at least two exactly once and returns exactly all MSCs.*

PROOF. Existence follows by reversing the repeated deterministic deletions in Theorem 4.2; every prefix is feasible and therefore passes the surplus test. Uniqueness follows because every non-root feasible clique has the single parent $C \setminus \{d(C)\}$, and only that parent accepts it as a child. The preceding containment argument proves that the final filter retains precisely the inclusion-wise maximal feasible cliques. $\square$

Complexity.

THEOREM 4.4 (TIME AND SPACE COMPLEXITY OF MSC-HIGHSIM). *Let F_τ be the number of feasible cliques of size at least two, K the maximum clique size, Δ the maximum degree, T the number of workers, and $N = |\mathcal{F}|$ the number of terminal candidates. A conservative*

bound for canonical search is

$$O\left(F_\tau \Delta(\Delta + K \log \Delta) + KN \log N + K \sum_{S \in \mathcal{F}} L_{\min}(S)\right),$$

with $O(m + T(n + K\Delta) + NK)$ words of space in addition to vector storage.

PROOF. Canonical ownership bounds accepted states by F_τ . A state tests at most Δ candidates. For each candidate, common-neighbor intersection and accumulator maintenance cost $O(\Delta)$ conservatively, while updating and comparing at most K member contributions uses cached similarities with $O(\log \Delta)$ lookup. This is bounded by $O(\Delta(\Delta + K \log \Delta))$ per accepted state. The final filter has the same $O(KN \log N + K \sum_S L_{\min}(S))$ cost as in MSC-BKPLUS.

Sparse similarity adjacency and neighbor bitmaps use $O(m)$ words. Each worker owns an $O(n)$ generation-tagged accumulator and $O(K\Delta)$ DFS scratch space. Terminal candidates occupy $O(NK)$ words; no feasible-size layer or visited-state table is materialized. $\square$

Canonical search is governed by the feasible-state family F_τ rather than structural maximal cliques. This family can exceed the structural output at intermediate thresholds but contracts sharply as τ becomes selective; MSC-BKPLUS instead continues to pay the structural MCE cost.

5 EXPERIMENTS

Eight research questions cover end-to-end efficiency (RQ1), threshold and graph-size scalability (RQ2–RQ3), parallelism and mechanisms (RQ4–RQ5), the MSC output space (RQ6), definition and pattern-family comparisons (RQ7), and a real publication case study (RQ8).

5.1 Experimental Setup

We compare against the following representative algorithms:

- **STRUCTBK** implements Bron–Kerbosch search with Tomita pivoting and degeneracy-ordered roots [5, 16, 17, 51]. Because it ignores semantic thresholds and enumerates structural maximal cliques, it provides a threshold-independent reference for structural search time and output size.
- **FastQC** enumerates maximal quasi-cliques satisfying a degree-density constraint [59]. We use it to compare MSCs with a widely used relaxed dense-subgraph family rather than as a same-definition runtime competitor.
- **FaPlex** enumerates maximal 2-plexes, in which each vertex may be nonadjacent to at most one other vertex [63]. It provides a second relaxed structural family for evaluating recovery, fidelity, density, and semantic coherence of the returned patterns.
- **MSC-BKBASIC** is the direct exact baseline developed in Section 3. It uses the same MSC definition, threshold, pairwise similarities, and strict-containment maximality criterion as MSC-BKPLUS and MSC-HIGHSIM, but retains pivot-free BK recursion and invokes a secondary CHECK search to validate semantic maximality.

Table 1: Datasets.

Dataset	$ V $	$ E $	D	Attributes
GitHub-Social	37,700	289,003	4,005	natural
ogbn-arxiv	169,343	1,157,799	128	natural
Flickr	89,249	449,794	500	natural
email-Enron	36,692	183,831	256	generated
sc-nasasrb	54,870	1,311,227	256	generated
sc-pkustk11	87,804	2,565,054	256	generated
sc-pwtk	217,891	5,653,221	256	generated
sc-lldoor	909,537	20,770,807	256	generated

- PAIRSIM-MCE removes every edge whose endpoint similarity is below τ and then applies structural maximal-clique enumeration. It represents the every-pair similarity requirement used by related constrained enumeration models [58]. We use it only as a definition-level result-family comparison to measure the group structure lost when aggregate coherence is replaced by a pairwise rule.

Datasets. The evaluation contains three graphs with natural attributes and five public graphs with reproducibly generated attributes. Flickr uses its official 500-dimensional user features [60]; GitHub-Social uses its official 4,005-dimensional developer features; and ogbn-arxiv retains its official 128-dimensional node features [24, 46]. For email-Enron and the four SuiteSparse graphs [13, 30], the PCG64 pseudorandom generator draws 256 coordinates independently from $U[-1, 1)$ under fixed dataset seeds, followed by ℓ_2 normalization. Every graph topology is extracted directly from its public source dataset. Generated vectors provide controlled inputs for runtime, scale, and mechanism experiments; the semantic-output analyses in RQ7 use only the three graphs with official natural attributes. Unless stated otherwise, subsequent multi-dataset analyses use the fixed panel of GitHub-Social, ogbn-arxiv, sc-nasasrb, and sc-lldoor. RQ3 instead uses nested subgraphs of ogbn-arxiv for controlled size scaling, and RQ7 uses the three graphs with observed attributes.

Protocol. Experiments run on an AMD Ryzen 9 7945HX workstation with 16 physical cores, 32 logical processors via simultaneous multithreading (SMT), and 32 GiB memory. We compile with GCC 13.2 and OpenMP. Each process is limited to 16 GiB and 32 OpenMP threads. The uniform timeout is 1,200 seconds. Completed runtime configurations are repeated three times and reported by the median; a configuration is marked time-limit exceeded (TLE) only if all three runs exceed the 1,200-second limit. Runtime and counter-instrumented runs are separated; end-to-end time covers the in-memory pipeline with optional text serialization disabled.

Absolute cosine similarities are not directly comparable across vector sources. We therefore calibrate the threshold separately for each dataset using the empirical distribution of cosine similarities over its edges. Specifically, q_p denotes the p th percentile of these edgewise similarities, and we set $\tau = q_p$. Thus, q_{20} is the similarity value that exceeds approximately 20% of adjacent node-pair similarities (equivalently, approximately 80% are at least q_{20}). All thresholds are computed and fixed before execution. The primary grid uses q_{20} , q_{80} , and q_{99} ; sensitivity experiments additionally evaluate q_5 , q_{50} , q_{70} , q_{90} , q_{95} , and q_{97} .

Table 2: [RQ1] Median end-to-end runtime (s) for structural, external, and exact MSC methods. TLE means all three runs exceed the 1,200-second limit.

Dataset	Structural / external			q	MSC methods		
	STRUCTBK	FastQC	FaPlex		BKBASIC	BKPLUS	HIGHSIM
GitHub-Social				20	34.777	0.327	29.315
				80	27.548	0.314	20.551
	0.110	5.74	4.28	99	21.495	0.391	0.118
ogbn-arxiv				20	1.590	1.233	3.357
				80	1.165	0.872	1.610
	0.309	35.26	28.43	99	1.249	0.843	0.197
Flickr				20	2.751	1.152	1.612
				80	2.910	0.760	0.599
	0.117	0.27	0.17	99	0.454	0.285	0.169
email-Enron				20	9.196	0.259	16.739
				80	8.312	1.266	0.083
	0.090	20.47	1.44	99	4.084	0.206	0.033
sc-nasasrb				20	TLE	0.301	TLE
				80	TLE	102.820	0.759
	0.090	7.10	67.30	99	TLE	0.415	0.079
sc-pkustk11				20	TLE	0.645	TLE
				80	TLE	TLE	2.219
	0.143	51.80	216.85	99	TLE	0.611	0.167
sc-pwtk				20	TLE	1.093	TLE
				80	TLE	TLE	3.977
	0.286	37.97	1190.44	99	TLE	1.364	0.421
sc-lldoor				20	TLE	4.870	TLE
				80	TLE	563.996	14.944
	1.679	267.83	TLE	99	TLE	4.845	1.658

5.2 RQ1: Exactness and End-to-End Efficiency

This experiment answers RQ1 by comparing complete MSC output sets and total runtime across the 24 fixed dataset-threshold workloads. Exhaustive oracles first test 225,000 vertex-bound instances and 40,001 canonical-parent instances, including negative similarities, equality at τ , and tied minimum contributions. On real graphs, jointly completed methods agree on normalized clique sets, counts, and size histograms. In particular, all three exact methods return 263,721, 71,118, and 2,636 MSCs on GitHub-Social at q_{20} , q_{80} , and q_{99} , respectively; MSC-BKPLUS and MSC-HIGHSIM also agree on every jointly completed row of Table 2. Across the eight graphs, the structural reference enumerates between 22,612 and 1,041,789 structural maximal cliques; these counts quantify the structural enumeration workload underlying the runtime comparisons. For the threshold-independent runtime references, FastQC uses $\gamma = 0.95$ with minimum size 10, while FaPlex uses $k = 2$ with minimum size 20; each parameterization is fixed across all eight graphs.

Table 2 first establishes the gain over the same-definition MSC-BKBASIC baseline. On every jointly completed row, the faster proposed method is $1.29\times$ – $182\times$ faster; MSC-BKBASIC additionally reaches TLE on all twelve workloads from the four larger generated-attribute graphs. MSC-BKPLUS completes every q_{20} workload and 22 of 24 rows, while MSC-HIGHSIM completes the two intermediate rows where host-clique subset enumeration reaches TLE. These results isolate the benefit of the two search organizations because all three methods enumerate exactly the same MSC family.

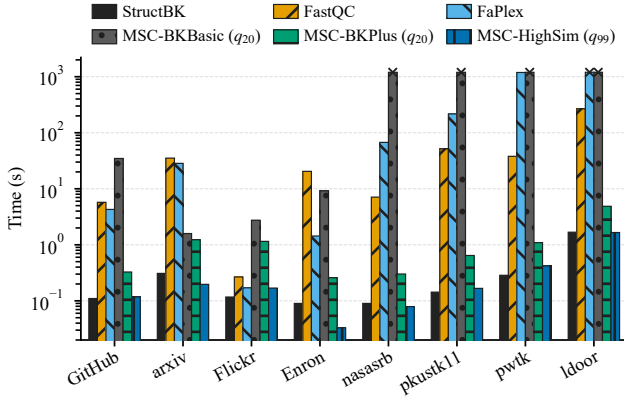


Figure 5: [RQ1] Runtime across structural, relaxed-family, and MSC methods. Crosses mark the 1,200-second limit.

Figure 5 then selects the operating point targeted by each organization: q_{20} for MSC-BKBasic and MSC-BKPlus, and q_{99} for MSC-HighSim. MSC-BKPlus completes all eight low-threshold workloads in 0.259–4.870 seconds; on seven graphs its runtime is within $4.6\times$ of threshold-independent structural MCE. MSC-HighSim completes all eight high-threshold workloads in 0.033–1.658 seconds, corresponding to $0.37\times$ – $1.47\times$ the structural runtime. Thus exact MSC enumeration approaches structural-enumeration efficiency in each method’s intended regime, despite the additional non-monotone semantic and maximality constraints. FastQC and FaPlex enumerate different relaxed pattern families, so their fixed parameter columns provide runtime context rather than same-output speedups.

5.3 RQ2: Threshold Sensitivity

Figure 6 applies a nine-point quantile grid to the fixed four graphs. MSC-BKPlus leads at low and intermediate thresholds, whereas MSC-HighSim takes over at q_{90} on GitHub-Social and q_{80} on Flickr. The generated graphs exhibit a larger intermediate output region: at q_{70} , sc-nasasrb and sc-ldoor produce 4.38 and 66.6 million MSCs. As q rises, infeasible insertions contract the canonical forest, reducing MSC-HighSim to 0.079 and 1.658 seconds at q_{99} . MSC-BKBasic is shown on the two natural-attribute graphs with complete curves; the specialized method remains faster in its favorable regime. These trajectories reflect the search designs: MSC-BKPlus amortizes decisions within structural hosts, whereas MSC-HighSim visits feasible states with canonical ownership.

5.4 RQ3: Scalability with Graph Size

Figure 7 reports absolute end-to-end runtime on shared logarithmic axes as the graph grows through nested induced ogbn-arxiv graphs whose vertices follow one fixed degree order. We use this single graph because it is both the largest observed semantic graph and supports a controlled nested expansion. The construction changes the number of vertices and outputs while preserving the graph family and all six threshold quantiles. Across the four sizes, at

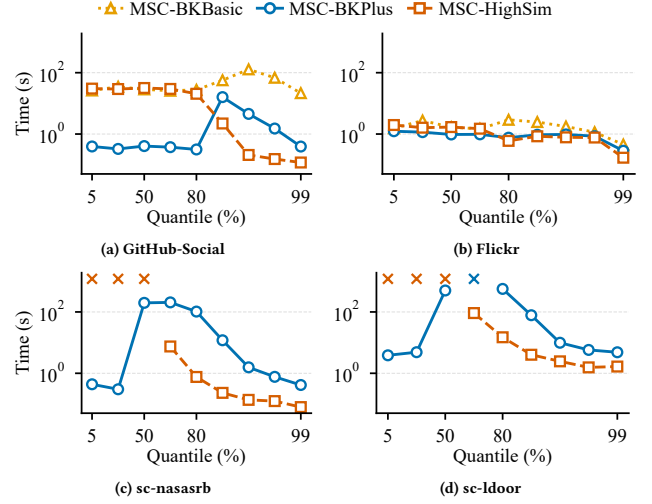


Figure 6: [RQ2] Increasing semantic selectivity reverses the runtime ordering of the exact search organizations.

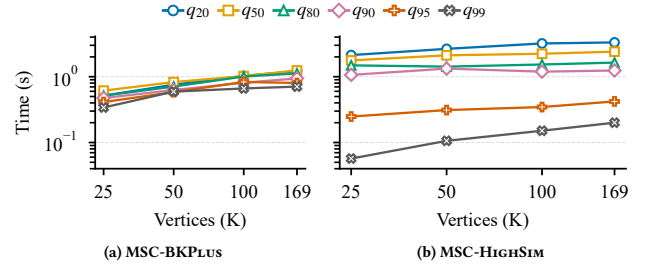


Figure 7: [RQ3] Both exact methods scale across graph sizes while the high-threshold crossover persists.

q_{20} , MSC-BKPlus increases from 0.516 to 1.216 seconds and MSC-HighSim from 2.130 to 3.320 seconds; at q_{99} , the corresponding increases are 0.340 to 0.707 and 0.057 to 0.200 seconds. The runtime ordering changes consistently at q_{95} , with MSC-BKPlus faster through q_{90} , while MSC-HighSim becomes faster at q_{95} and q_{99} . The latter’s q_{99} advantage is $6.0\times$, $5.6\times$, $4.4\times$, and $3.5\times$ at 25K, 50K, 100K, and 169K vertices, respectively, so the selective-threshold benefit persists as the graph grows by $6.8\times$. Thus, the high-selectivity advantage persists under controlled graph growth rather than arising from one graph size.

5.5 RQ4: Parallel Scalability

Figure 8 evaluates 1–32 threads at q_{20} for MSC-BKBasic and MSC-BKPlus and q_{99} for MSC-HighSim. Speedups are computed per graph before their geometric mean. The proposed methods use four graphs; MSC-BKBasic uses the two completing natural-attribute graphs. Their one-thread ranges are 1.054–13.044, 0.140–2.443, and 13.841–84.999 seconds for MSC-BKPlus, MSC-HighSim, and MSC-BKBasic, respectively.

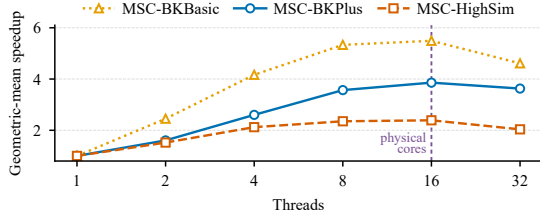


Figure 8: [RQ4] Root-level scheduling accelerates all three exact MSC searches.

At 16 threads, MSC-BKBasic, MSC-BKPlus, and MSC-HighSim reach 5.49 $\times$, 3.86 $\times$, and 2.39 $\times$ speedup, respectively. Longer MSC-BKBasic searches amortize scheduling and merging most effectively. Host roots in MSC-BKPlus expose more parallel work, whereas MSC-HighSim’s selective canonical forests finish quickly and leave less work per root. All curves peak at 16 physical cores; 32 SMT threads add shared-resource contention without adding cores.

5.6 RQ5: Mechanism Ablations

Table 3 changes one mechanism at a time; completed variants return identical MSC sets, and instrumented counters are excluded from timing. Ablations use q_{20} except for BKPlus’s q_{99} vertex-bound row and HighSim’s q_{90} state-generation rows. In MSC-BKPlus, sorted-edge and local-containment pruning reduce subset tests by 1.08 $\times$ and 1.51 $\times$. Removing the vertex bound increases tests by 1.27 $\times$ –6.20 $\times$ on three graphs; on sc-lldoor, the full method takes 6.04 seconds while the ablation exceeds 30 seconds. In MSC-HighSim, canonical ownership reduces candidate tests by 1.50 $\times$ and avoids 18,942 layered duplicate states. Against Visited DFS and Layered traversal at q_{90} , it is 4.38 $\times$ and 3.64 $\times$ faster and uses 1.73 $\times$ and 1.53 $\times$ less peak memory. The memory difference is largest when duplicate-state storage grows beyond the shared graph and vector representation.

Sparse rows and direct marginal sums improve MSC-BKBasic by 1.12 $\times$ and 1.39 $\times$; local containment improves MSC-BKPlus by 1.19 $\times$.

Table 3: [RQ5] Search and engineering ablations; each row changes only the stated mechanism.

Method	Mechanism / comparison	Metric	Effect
<i>(a) Semantic pruning</i>			
BKPlus	Sorted-edge bound	tests / 4-graph GM	1.08 $\times$ fewer
BKPlus	Vertex-contribution bound	tests / 4 graphs	1.27–6.20 $\times$; one > 30 s
BKPlus	Local containment	tests / 4-graph GM	1.51 $\times$ fewer
<i>(b) State generation</i>			
HighSim	Canonical / Visited DFS	candidate tests / 4-graph GM	1.50 $\times$ fewer
HighSim	Canonical / Layered	duplicates / 4-graph sum	18,942 avoided
HighSim	Canonical / Visited DFS	time; peak GiB / 4-graph GM	4.38 $\times$; 1.73 $\times$
HighSim	Canonical / Layered	time; peak GiB / 4-graph GM	3.64 $\times$; 1.53 $\times$
<i>(c) Engineering choices</i>			
BKBasic	Sparse rows	runtime / GM*	1.12 $\times$ faster
BKBasic	Direct sum	runtime / GM*	1.39 $\times$ faster
BKPlus	Local containment	runtime / 4-graph GM	1.19 $\times$ faster

*Geometric mean over GitHub-Social and ogbn-arxiv, where complete BKBasic runs are available.

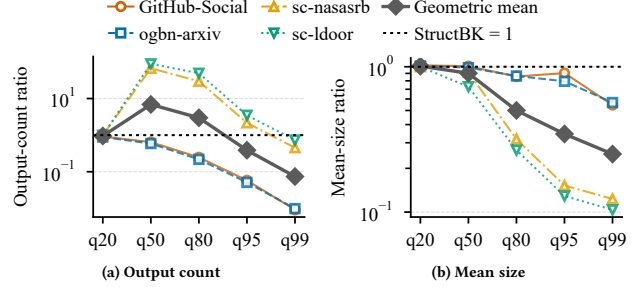


Figure 9: [RQ6] MSC output expands at intermediate thresholds and contracts under high selectivity.

5.7 RQ6: MSC Output Space

This experiment measures size-at-least-two output counts and mean sizes on four graphs: GitHub-Social, ogbn-arxiv, sc-nasasrb, and sc-lldoor. It uses five quantiles from q_{20} through q_{99} . Figure 9 normalizes each value by StructBK on the same graph. The geometric-mean count ratios are 0.95, 6.89, 2.97, 0.39, and 0.07: intermediate thresholds split large structural cliques into many feasible subsets that are maximal by inclusion. Selective thresholds instead contract the result family. Mean size decreases from 1.01 to 0.25 of the structural reference over the same grid. The count peak and size decline describe the non-monotone search difficulty more precisely than either statistic alone: one structural host can yield many smaller MSCs in the intermediate regime, whereas high thresholds eliminate most extensions and leave fewer, smaller outputs. This expansion–contraction pattern explains the threshold-dependent runtime regimes observed in RQ2.

5.8 RQ7: Group Preservation and External Pattern Families

Because this experiment evaluates semantic group preservation, it uses the three graphs with observed vector attributes: GitHub-Social, ogbn-arxiv, and Flickr. The reference cover contains all size-at-least-ten q_{80} MSCs. PairSim uses the same threshold, FastQC uses $\gamma = 0.95$ and minimum size 10, and FaPlex uses $k = 2$ and minimum size 20. We evaluate every baseline with Extended B³, an overlap-aware entity-level measure [1]. For each vertex, it measures how many reference co-members remain grouped with that vertex (recall) and how many reported co-members are supported by the reference cover (precision), then averages over vertices. This construction handles overlapping result families without imposing a one-to-one group matching; MSC self-comparison equals one on both axes.

Figure 10 exposes complementary losses under one metric. PairSim attains high precision on GitHub-Social and Flickr (0.920 and 0.907), but recall is only 0.156 and 0.169 because one weak pair fragments a collectively feasible group. StructBK and FastQC recover more MSC co-memberships, reaching recall 0.988 on ogbn-arxiv, while their precision on that graph is 0.626 and 0.598 because structural or relaxed maximality admits additional group relations. FaPlex is highly precise but, under the fixed minimum-size-20 protocol, recalls only 0.041–0.379. Thus none of the four alternative result families preserves both aspects of the MSC cover.

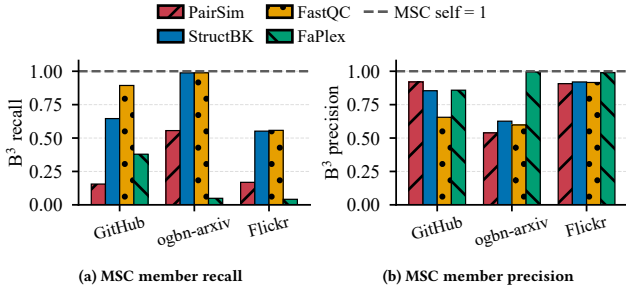


Figure 10: [RQ7] Alternative pattern families lose complementary aspects of MSC group structure.

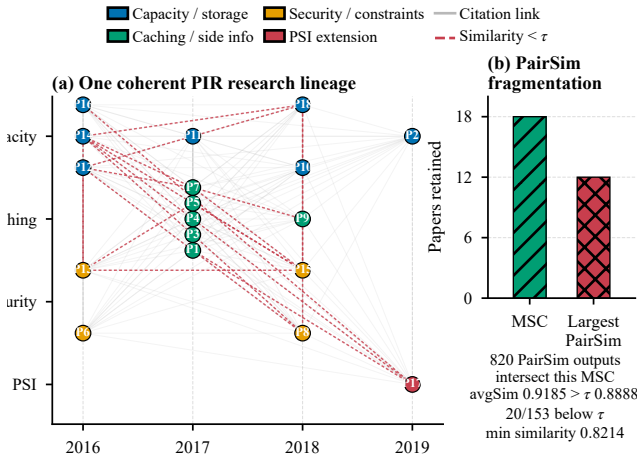


Figure 11: [RQ8] MSC recovers one 18-paper PIR research lineage that PairSim disperses across hundreds of overlap outputs.

Table 4: One representative publication from each thematic strand in the PIR MSC.

Theme in Figure 11	Representative publication
Capacity / storage	P14: <i>The Capacity of Private Information Retrieval</i> (Sun and Jafar, 2016; arXiv:1602.09134)
Caching / side information	P3: <i>Fundamental Limits of Cache-Aided Private Information Retrieval with Unknown and Uncoded Prefetching</i> (Wei, Banawan, and Ulukus, 2017; arXiv:1709.01056)
Security / constraints	P8: <i>Private Information Retrieval Through Wiretap Channel II: Privacy Meets Security</i> (Banawan and Ulukus, 2018; arXiv:1801.06171)
PSI extension	P17: <i>Private Set Intersection: A Multi-Message Symmetric PIR Perspective</i> (Wang, Banawan, and Ulukus, 2019; arXiv:1912.13501)

5.9 RQ8: Case Study

We examine the largest label-consistent q_{80} MSC containing a below-threshold pair on ogbn-arxiv. The OGB mapping resolves its 18 vertices to 2016–2019 cs.IT publications forming a structural clique [24]. Their titles connect PIR capacity, coded and cache-aided storage, side information, colluding and wiretapped databases, asymmetric traffic, and private set intersection. Table 4 gives one representative publication per thematic strand in Figure 11.

At $\tau = 0.8888$, the group has $\text{avgSim} = 0.9185$, although 20 of 153 pairs fall below τ (minimum 0.8214). Among 820 intersecting PairSim outputs, the best retains only 12 papers. MSC instead yields one reviewable PIR lineage for topic overview and expert discovery.

6 RELATED WORK

Clique Enumeration and Listing. Exact MCE builds on BK backtracking and pivoting [5, 51], output bounds [6, 39], polynomial-delay and output-sensitive enumeration [37, 52], and sparse-graph ordering [7, 16, 17]. Recent work reduces search subproblems or summarizes clique families [14, 34, 55], while large-graph implementations exploit specialized structures and parallelism [8, 12, 31]. Related variants address maximum-clique optimization [45], uncertain-graph MCE [11], signed-clique search [32], and fixed-size k -clique listing [54]; none enumerates inclusion-wise maximal cliques under an aggregate vector constraint.

Constrained Cohesive Subgraphs. FastQC [59] and FaPlex [63] enumerate maximal quasi-cliques and k -plexes, biplex methods relax bipartite completeness [10], and maximum k -plex search returns one optimum [22]. Pair-thresholded similarity cliques and weighted short-text cliques use different semantic contracts [19, 25]; fairness-aware methods constrain categorical balance [42, 61]. Similar and similar-weight bicliques impose hereditary pairwise or edge-weight conditions in bipartite graphs [57, 58]. In contrast, MSC retains a complete general graph, enumerates every inclusion-wise maximal solution, and applies a non-hereditary average over real-valued vectors.

Attributed Community Search. Classical community models motivate cohesive grouping through minimum-degree structure, modularity, and density [21, 41, 48]; attributed-community surveys further distinguish how topology and attributes are fused [4, 9]. Representative methods return partitions [47, 49, 56, 62], focused clusters or outliers [43], and query-specific influential, attributed, or inductive communities [18, 20, 26, 33, 53]. Subspace objectives [23, 40], directed truss search [35], and directed densest-subgraph discovery [36] likewise relax completeness or return partition/query/optimization results rather than all maximal complete groups under an aggregate threshold.

Vector Attributes. Representation models [3, 29, 44, 50] can supply the vectors, but approximate vector retrieval addresses a separate nearest-neighbor task [2, 27, 38]; MSC instead evaluates its aggregate threshold exactly on fixed attributes.

7 CONCLUSION

We formulated exact MSC enumeration under average pairwise similarity and proved its size decision problem NP-complete. MSC-BKPLUS combines structural decomposition with admissible semantic bounds; MSC-HIGHSIM turns feasible minimum-contribution deletion into a canonical reverse-search forest. Their complementary threshold regimes jointly complete all 24 fixed dataset–threshold workloads, with MSC-HIGHSIM $1.69\times$ – $6.24\times$ faster than MSC-BKPLUS on every q_{99} row. Ablations attribute these gains to semantic pruning and canonical-state contraction; natural-attribute results show that averaging preserves complete groups fragmented by pairwise thresholding.

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